Andrea P. Hernández Guzmán

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EDUCATION

Ph.D. in Industrial Engineering — 2024–Present
Georgia Institute of Technology, Atlanta, GA — Expected May 2029
Advisor: Dr. Nagi Gebraeel
Research: Machine learning for fault diagnostics under uncertainty and data privacy.

B.S. in Industrial Engineering — 2024
University of Puerto Rico – Mayagüez, PR
GPA: 3.87/4.00, *Magna Cum Laude*

RESEARCH EXPERIENCE

Graduate Research Assistant — Aug 2024–Present
Predictive Analytics and Intelligent Systems (PAIS) Lab, Georgia Institute of Technology, Atlanta, GA

- Designed a federated learning framework that enables multiple organizations to jointly diagnose equipment faults without sharing their sensitive operational data.
- Developed a distributionally robust multiclass fault-diagnostic model that remains reliable under uncertainty in both sensor features and fault labels.
- Integrated differential privacy into the framework to provide formal privacy guarantees for each participating organization, developing a new reformulation of the model and an algorithm to solve it.
- Implemented the complete framework in Python and Gurobi. Presented results at the IISE Annual Conference 2026, with a manuscript in preparation.

Undergraduate Research Fellow — Summer 2022
NSF REU Site, Purdue University, West Lafayette, IN

- Evaluated peer-to-peer (P2P) energy trading as an alternative to net-metering policies for commercial solar energy users.
- Built a Python simulation of energy transaction bidding within a microgrid to compare the economic and technical performance of both mechanisms.
- Published the results in *Eng* (2023) and presented them at Purdue's Summer Undergraduate Research Symposium and the NSF EEC Grantees Conference.

PUBLICATIONS

Manuscripts in Preparation

1. **Hernández-Guzmán, A.**, Ibrahim, M., & Gebraeel, N. Federated Learning for Multiclass Fault Diagnosis with Distributional Robustness and Differential Privacy. *In preparation.*

Peer-Reviewed Journal Articles

2. Soto, E. A., Wollega, E., Ortega, A. V., **Hernandez-Guzman, A.**, & Bosman, L. (2024). Reduction in emissions by massive solar plant integration in the US power grid. *Energies*, *17*(7), 1611. https://doi.org/10.3390/en17071611
3. Soto, E. A., Ortega, A. V., **Hernandez-Guzman, A.**, & Bosman, L. (2023). Commercial level analysis of P2P vs. net-metering comparing economic and technical indexes. *Eng*, *4*(3), 2254–2273. https://doi.org/10.3390/eng4030129
4. Soto, E. A., **Hernandez-Guzman, A.**, Vizcarrondo-Ortega, A., McNealey, A., & Bosman, L. (2022). Solar energy implementation for health-care facilities in developing and underdeveloped countries: Overview, opportunities, and challenges. *Energies*, *15*(22), 8602. https://doi.org/10.3390/en15228602

PRESENTATIONS

1. **Hernández-Guzmán, A**., Ibrahim, M., & Gebraeel, N. (November 2026). "A Distributionally Robust and Differentially Private Framework for Federated Multiclass Classification." Oral presentation at the INFORMS Annual Meeting, San Francisco, CA. *(Upcoming)*
2. **Hernández-Guzmán, A**., Ibrahim, M., & Gebraeel, N. (May 2026). "A Distributionally Robust and Differentially Private Framework for Federated Multiclass Classification." Oral presentation at the IISE Annual Conference & Expo, Arlington, TX.
3. **Hernández-Guzmán, A**. (September 2024). "Commercial Level Analysis of P2P vs. Net-Metering Comparing Economic and Technical Indexes." Poster presentation at the NSF EEC Grantees Conference, Alexandria, VA.
4. **Hernández-Guzmán, A**. (July 2022). "Commercial Level Analysis of P2P vs. Net-Metering Comparing Economic and Technical Indexes." Oral presentation at the Summer Undergraduate Research Symposium, Purdue University, West Lafayette, IN.

FELLOWSHIPS AND AWARDS

National GEM Consortium Ph.D. Fellowship — 2024–Present
Competitive national fellowship supporting graduate study in engineering; sponsored by NASA.
HACU Capital One HSI Immersion Program — 2023
Selected through a competitive application and interview process.
Academic Excellence Scholarship, University of Puerto Rico – Mayagüez — 2020–2024
Awarded annually for maintaining a high GPA.

PROFESSIONAL EXPERIENCE

Supply Chain Intern — Summer 2024
NASA Ames Research Center, Moffett Field, CA

- Collected U.S. and international import/export data from global trade platforms to trace where each part used in U.S. drone assembly is sourced, back to the start of the supply chain.
- Used this data to build geocoding and interactive map features for NASA's Advanced Air Mobility (AAM) tool, letting users visualize drone suppliers and part flows by location.
- Authored a report on the U.S. drone market, examining why domestic production falls short of national demand and drives reliance on Chinese-made drones.

Data Analyst Intern — Summer 2023
Verizon, Remote

- Investigated data integrity gaps between Verizon's network equipment database (CND) and other internal systems, caused by rising project volume and a newly introduced work-order process.
- Built network equipment records and work-orders across multiple projects and business units, working directly with Edge Implementation engineers to identify where the process was breaking down.
- Redesigned the CND entry and reporting process, reducing completion time by over 40%, and presented the new process flow and automation recommendations to engineering leadership.

MENTORING AND OUTREACH

Graduate Mentor — Fall 2025
Community Service Project, Center for Academics, Success, and Engagement, Georgia Institute of Technology

- Mentored 2 mixed-year teams of 10 students each through 10-week projects with community partners, guiding their problem solving, teamwork, and decision making.
- Guided one team in automating a nonprofit's donation and request process, replacing a manual email-based system.
- Guided a second team, in partnership with Invest Atlanta, in building an interactive map that identified gaps in food bank coverage across Atlanta communities.

PROFESSIONAL MEMBERSHIPS

Institute of Industrial and Systems Engineers (IISE) — 2026–Present
Institute for Operations Research and the Management Sciences (INFORMS) — 2025–Present
Alpha Pi Mu – Industrial Engineering Honor Society — 2022–Present

- Community Activities Coordinator, UPR–Mayagüez Chapter, 2023–2024

Tau Beta Pi – The Engineering Honor Society — 2022–Present

SKILLS

Programming: Python, SQL, MATLAB
Optimization & Statistics: Gurobi, Minitab
Languages: Spanish (native), English (fluent)